

Neel Maddu

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SUMMARY

Computer Science student at Georgia Tech with a strong focus on artificial intelligence, systems, and mobile development. Currently a Software Engineer Intern at GTRI's CIPHER Lab, applying ML and systems architecture to cybersecurity workflows. As a Tech Lead for the Georgia Tech iOS Club, I co-direct a 20+ member cross-functional team to ship production-level iOS applications leveraging Swift, SwiftUI, and Core ML.

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science | **GPA: 4.0**

Atlanta, GA

Expected May 2027

EXPERIENCE

Georgia Tech Research Institute (GTRI)

May. 2026 – Present

Software Engineer Intern, CIPHER Lab – Threat Analysis Branch

Atlanta, GA

- Developing ML Research & Development solutions and data analytic pipelines in support of cybersecurity and intelligence analysis workflows.
- Applying NLP techniques to analyze and extract insights from large-scale unstructured data in a national security context
- Evaluating the risk posture of LLM-enabled and agentic systems within secure experimental frameworks to determine how architectural choices impact resilience against emerging threats.
- Translating technical evaluations of AI-model vulnerabilities into high-level insights to inform security evaluation and decision-making.

Georgia Institute of Technology – VIP Program

Jan. 2026 – Present

Automated Algorithm Design – AlphaEvolve Subteam

Atlanta, GA

- Contribute to a GTRI-sponsored research program developing a multi-objective genetic programming framework that automatically evolves hybrid algorithms from data, outperforming human-designed baselines.
- Work on the AlphaEvolve subteam extending the LLM-Guided Evolution (LLM-GE) framework — cited as prior work by Google DeepMind's AlphaEvolve — which uses LLMs to propose, mutate, and evaluate code in an automated evolutionary loop.
- Run and validate experiments on Georgia Tech's PACE ICE HPC cluster, developing reproducible Python workflows for neural architecture search and performance benchmarking.

Georgia Tech iOS Club

Aug. 2025 – Present

Tech Lead

Atlanta, GA

- Co-lead a 20-member project team, collaborating alongside another Tech Lead and 4 Senior Developers to architect and ship production-level iOS applications using Swift, SwiftUI, and Firebase.
- Architect application features using the MVVM design pattern, ensuring a scalable and maintainable codebase while streamlining the onboarding process for team members.
- Conduct code reviews and manage GitHub pull requests, providing continuous technical mentorship to enforce code quality and mobile development best practices.

PROJECTS

Multimodal AI Search Engine | PyTorch, FAISS, NumPy, pandas, Streamlit, Python

Feb. 2025 – Nov. 2025

- Developed a visual search engine by indexing over 10,000 images using CLIP embeddings and FAISS, retrieving semantically relevant images from natural language prompts with an average query time of 2 seconds.
- Built and deployed a Streamlit interface for real-time, scrollable image results, demonstrating practical ML deployment and cross-modal embedding techniques.

JotDown | Swift, SwiftUI, Git, GitHub, Firebase, ML

Sep. 2025 – Dec. 2025

- Architected the application's UI with SwiftUI, employing an MVVM design pattern to build a responsive and minimalist interface for frictionless data entry.
- Integrated on-device Core ML models to perform Natural Language Processing, enabling a semantic search feature that recognizes entities like dates and keywords within user queries.
- Designed a data visualization module to generate a knowledge graph from user notes, modeling semantic relationships to uncover latent themes and connections.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, FAISS, MediaPipe, NLP

Data: pandas, NumPy, Graph Analytics

iOS Development: Swift, SwiftUI, Core ML, MVVM

Backend & Databases: SQL, MySQL, MongoDB, Firebase, Node.js

Programming Languages: Python, Java, C, JavaScript, C++, Next.js, React

Tools: Git, GitHub, CI/CD, Jenkins, GitHub Actions, Docker

Cloud: AWS, Azure